

Research on User Transparency Perception Based on Mixed Reality

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Abstract—The user's visual perception in mixed reality (MR) includes the combination of the real physical environment and virtual objects rendered in an optical see-through head mounted display (OST-HMD), so the presentation of MR rendering objects will be affected by the light from the real world; thus, rendering effect deviates from the original design intent. This paper focuses on the user's perception of transparency in mixed reality and the impact of ambient illumination on the perception of transparency in mixed reality. We conducted a psychophysical experimental study and recruited 20 participants to recognize the transparency of virtual objects under two kinds of ambient illumination via Microsoft HoloLens and select the corresponding value in the slider. Our analysis shows that the transparency of virtual objects will be underestimated when the ambient illumination is high. Additionally, the experiments also found that as the magnitude of transparency increases, the perception of transparency will be underestimated. This study can provide a reference for the design of MR interface transparency coding.

Keywords—transparency, visual perception, mixed reality, illumination

I. INTRODUCTION

Mixed Reality (MR), as an iteration of Virtual Reality (VR) and Augmented Reality (AR), organically combines physical and virtual environments (VE), resulting in a new way of visualizing the environment, and providing users with a more natural interactive environment. Through the spatial computing of MR, inputting information into our surroundings, we can redesign and interpret our physical world in unprecedented ways [1]. The second generation of Microsoft HoloLens released by Microsoft in 2019 is a representative device of mixed reality. HoloLens uses near-eye 3D diffractive display technology, equipped with two waveguides see-through holographic lenses, whose virtual content adopts digital light processing (DLP) projection technology, projected from the front micro projector across the waveguide lenses to the human eye, and it also allows the light of the real world to penetrate. Compared with its first-generation product, it has obvious improvements in optical components, depth perception, eye tracking, gesture tracking, etc., presenting a more pleasant interactive experience for users. As a different form of visual presentation, mixed reality technology provides more possibilities for the presentation and communication of information, and is beginning to play a role in many fields (including healthcare, architecture and civil engineering, manufacturing, defense, tourism, automation and education), gradually changing the way these fields work [2].

In AR and MR, simultaneous visualization of real environments and virtual objects is critical. Visual variables are the visual elements that constitute graphic symbols, also known as graphic elements or visual contrast elements. In visual design, it is through the changes of these visual variables to express the difference of data information [3].

Bertin [4], one of the founders of data visualization, defined seven basic visual variables for vision, including position variables and six retinal variables, namely size, shape, brightness, hue, orientation, and texture, laying the foundation for visualization research. Some later scholars supplemented the visual variables on this basis - Joel Morrison [5] supplemented saturation and layout; and Alan MacEachren [6] supplemented blurriness, resolution and transparency. Among them, transparency is not an attribute of color, but it is a visual attribute that affects color perception [7]. Transparency refers to the degree to which an object allows light to pass through. The use of visual variables to encode information is a research focus in visualization. For the visualization of spatiotemporal data in a three-dimensional environment, the encoding form of transparency not only has a semantic mapping of the importance level, but can also reflect the gradual changes in the time dimension [8]. Some researchers tried to use transparency in campus multiscale temporal information visualization [9] and 3D cadastral visualization [10]. On the other hand, the use of transparency in MR can also solve the occlusion problem in MR [11]. In addition, transparency can also affect depth perception, so as to enhance the sense of depth [12].

In terms of the information presentation form, the research of Zhang and Murdoch [13] showed that there are different transparency perception characteristics between optical see-through augmented reality and traditional liquid crystal display (LCD). Compared with the traditional two-dimensional visualization, the 3D presentation method increases the depth extension, and the display of the existing MR equipment and the traditional two-dimensional interface is also fundamentally different. Some studies have shown that 3D presentation can bring visual clutter and bring additional cognitive complexity to users [14][15][16]. On the other hand, 3D visualization has indispensable advantages in the more comprehensive information processing of the target, such as recognizing phenomena, naming objects, point recognition or scene interpretation [1]. Therefore, it is very important to study the perceptual characteristics of visual variables in mixed reality, that is, the differences in the perception of transparency in different visualization environments.

This paper studies the characteristics of people's perception of transparency under different lighting environments in a mixed reality environment, and provides method guidance for immersive visualization design, to present a more efficient and more human cognitive characteristic of the visual analysis environment.

II. EXPERIMENTAL STUDY

We conducted a psychophysics experiment to study the influence of ambient illumination levels on the perceptual bias and accuracy of transparency in mixed reality, to verify the experimental hypothesis that the transparency of virtual

objects will be underestimated when the ambient illumination is high.

This experiment uses scaling and psychophysical methods to measure the perceptual bias of people's transparency in mixed reality. We improved the experimental method of Zhang and Murdoch[13] and applied it to mixed reality. The subjects were asked to evaluate the perceived transparency of the test squares by moving the slider. The slider range was 0-100%, and the step size was 1%.

In our experiment, white squares with the same hue ($R=255, G=255, B=255$) with different transparencies will be rendered into HoloLens in turn. Subjects were asked to look at the square and express the perceived transparency in the slider below. A program on unity records data about the participant's slider operations. The perceptual bias of transparency is calculated by subtracting physical transparency from perceived transparency. This section describes the experimental apparatus, environment, experimental tasks, and experimental procedures in detail.

A. Apparatus and Environment

This experiment was carried out in Microsoft HoloLens 2, and the device brightness was adjusted to the highest value during the experiment. The ambient illuminance is controlled by artificial light sources and measured by a VICTOR 1010C digital illuminance meter. The MR scene is built by Unity 3D v2019.4.36f1c1, and the interactive controls are selected and modified from the HoloLens Mixed Reality Toolkit. The white square was located in an $8^\circ \times 8^\circ$ area in the center of the background, with a side length of 10 cm, 70 cm from the subject's eyes, and placed in front of a sinusoidal grating background in a physical environment. Since it is difficult for users to make decisions about transparency on a low-contrast background, a sinusoidal grating was used as the background in this experiment to improve the contrast of the background.

B. Participants

In China, we recruited 20 students from the School of Mechanical Engineering of Southeast University to participate in the experimental study, including 10 men and 10 women aged between 22 and 25 years ($M=24.05, SD=0.97$). They are all graduate students of industrial design or mechanical engineering. All subjects had normal or corrected-to-normal vision, and had no visual diseases such as color blindness. Three participants have no experience with extended reality (XR), six participants have used XR equipment (less than two hours), and 11 participants are experienced users (more than two hours).

C. Variables

This experiment involves two independent variables - ambient illuminance and transparency level. Two ambient illuminances were examined in this experiment. Ten transparency levels of test patches were examined for each ambient illuminance. Each level was repeated five times. In total, 2 ambient illuminance $\times$ 10 transparency levels $\times$ 5 repetitions resulted in 100 trials per observer.

Ambient illuminance. To evaluate the impact of physical ambient illuminance on users' perception of transparency in MR, we control the lighting in the environment and set 2 light levels, 10-20 lux (lx) in a dark room and 400-500 lux (lx) in a bright room. The experimental order among subjects was balanced by the ABBA method to balance the effects of practice and fatigue.

Transparency level. Considering the impact of different levels of transparency on perception, 10 transparency levels were set in the experiment, namely 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% and 100%. Five repetitions were performed for each level. The trial order was sorted by Latin square to balance the order effect and avoid expectation error.

D. Procedure

Before the experiment, the participants were told the experimental task. Participants wore HoloLens glasses and adjusted for calibration. After calibration, the participants practiced the operation of the HoloLens interactive gestures, and then previewed the transparency effects corresponding to different transparency values, so that the subjects were pre-familiar with the rendering characteristics of HoloLens, and then conducted simulated experimental exercises to ensure that the participants were clear about the experimental tasks and how to use the device.

Afterwards, the participants began the formal experiment: a white square with a slider and a blue button appeared in the field of view, as shown in Figure 1. Participants first assessed the transparency of the white square that appeared, then expressed the observed transparency value by moving a slider between the two endpoints, and then clicked the next button to proceed to the next trial, repeating the previous operation. A "+" sign appears for 500 milliseconds before the next trial to minimize the effect of visual afterimages, as shown in Figure 2. Participants had unlimited time to evaluate before making a decision. All test squares were checked in this light condition before moving on to the next light condition, and each square was repeated 5 times.



Fig. 1. Experimental scene. Left: participant observation perspective. Right: third-party perspective.

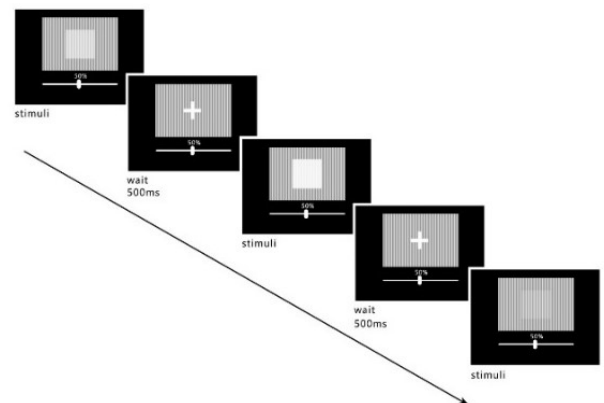


Fig. 2. Experimental flow chart

III. RESULT

During the experiment, we collected the values that participants chose on the slider for each trial and compared them with the corresponding physical transparency settings.

First, the perceptual bias for each trial was calculated, which was calculated by subtracting the physical transparency (that is the transparency set by unity3D) from the perceived transparency (that is the value selected by the subjects on the slider). The data distortion (approximately 0.4%) caused by misoperation was eliminated.

A. Descriptive Statistics

We calculated the mean and standard deviation of perceptual bias for 5 (repetitions) $\times$ 20 (participants) trials under 2 (ambient illumination) $\times$ 10 (transparency levels) experimental conditions, as shown in Table I. The mean represents the perceptual bias, in which a positive value indicates that the perceptual value is higher than the actual value, while a negative value indicates that the perceptual value is lower than the actual value. The standard deviation represents the perceptual stability. As shown in Figure 3, the mean of perceptual deviation in a high illumination environment is lower than that in a low illumination environment, so the experimental hypothesis is verified, that is, the transparency of virtual objects will be underestimated when the ambient illumination is high.

TABLE I. MEAN (M) AND STANDARD DEVIATION (SD) OF PERCEPTUAL BIAS FOR EACH EXPERIMENTAL CONDITION

Trans- parency	Low illumination		High illumination	
	M	SD	M	SD
10%	2.62%	8.71%	-2.13%	6.49%
20%	2.34%	13.41%	-2.85%	11.08%
30%	-0.12%	11.38%	-3.66%	14.41%
40%	-3.98%	12.62%	-5.25%	15.89%
50%	-5.27%	14.81%	-8.29%	15.67%
60%	-5.98%	14.31%	-9.18%	17.30%
70%	-6.85%	15.46%	-10.72%	20.94%
80%	-7.30%	15.04%	-11.47%	19.60%
90%	-9.66%	13.98%	-15.13%	18.53%
100%	-15.06%	15.09%	-20.18%	17.24%

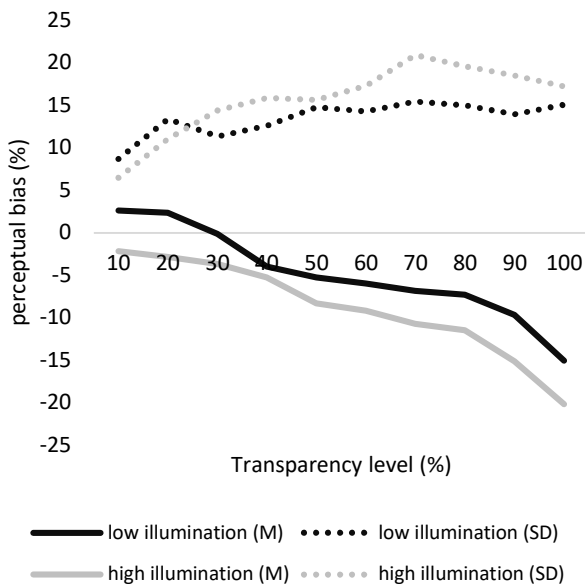


Fig. 3. Mean (M) and standard deviation (SD) of perceptual bias for each experimental condition

B. Variance Analysis

The results of Mauchly's test of sphericity were significant ($p=0.000<0.05$), which did not conform to the spherical distribution, so the analysis of variance used the results of the within-subject effect test. The analysis showed that the main effect of environmental illuminance was significant ($F(1,90)=11.229$, $p=0.001<0.05$). When the ambient illumination is high, the transparency of virtual objects will be underestimated, which verifies the experimental hypothesis. The main effect of transparency level was significant ($F(4.641, 417.681)=37.630$, $p=0.000<0.05$). As transparency increases, the perception of transparency will be underestimated. The within-subject comparison test showed that the downward trend was in line with a linear function ($F(1, 91)=100.284$, $p=0.000<0.01$). The interaction between ambient illuminance and transparency level was not significant ($F(6.599, 593.884)=0.537$, $p=0.774>0.05$). Perceptual bias has a similar decreasing trend in low-light and high-light environments. There was no significant difference in the experimental results between participants of different genders ($F(1,90)=0.448$, $p=0.505>0.05$).

IV. DISCUSSION

In this study, we investigate the effect of ambient light on the perception of the transparency of virtual objects in HoloLens and carry out psychophysical experimental research. The experimental hypothesis is confirmed by experimental studies; that is, the transparency of virtual objects will be underestimated when the ambient illumination is high. Under the condition that the brightness of the virtual objects rendered by HoloLens is constant, when the ambient illumination increases, the relative brightness of the virtual objects will be weakened, so the participants' perception of transparency is low. Therefore, in the process of mixed reality design and development, the ambient illumination of the usage scene should be considered; for example, the device brightness can be adjusted adaptively according to the light. Additionally, the experiments also found that as the magnitude of transparency increases, the perception of transparency will be underestimated. The rendering mechanism of HoloLens is different from traditional two-dimensional screens and cannot achieve a completely opaque rendering effect; that is, it cannot completely cover the real environment. In addition, during the experiment, participants reported that in addition to the change in transparency, there was also a significant change in brightness in the interval with greater transparency, so transparency and brightness still have a certain similarity in mixed reality [8].

In this experiment, considering the duration of the experiment, the effect of color on the perception of transparency is not considered. In addition, the change in shape may also have a certain impact on the perception of transparency. In mixed reality, in addition to two-dimensional objects, there are three-dimensional objects, such as cubes and spheres. Whether the perception of transparency is consistent under different hues or shapes will be the focus of future research.

V. CONCLUSION

In this study, we investigate the effect of ambient illumination on the perception of the transparency of virtual objects in HoloLens. Psychophysical experimental research was carried out. The experimental hypothesis is confirmed by experimental studies, that is, the transparency of virtual

objects will be underestimated when the ambient illumination is high. Additionally, the experiments also found that as the magnitude of transparency increases, the perception of transparency will be underestimated. In future work, we will try to study the consistency and stability of transparency perception under different hues and different shapes.

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REFERENCES

- [1] Çöltekin A, Lochhead I, Madden M, Christophe S, Devaux A, Pettit C, Lock O, Shukla S, Herman L, Stachoň Z, Kubiček P, Snopková D, Bernardes S, Hedley N. Extended Reality in Spatial Sciences: A Review of Research Challenges and Future Directions[J]. *International Journal of Geo-Information*, 2020, 9(7):439.
- [2] Park S, Bokijonov S, Choi Y. Review of Microsoft HoloLens Applications over the Past Five Years. *Applied Sciences*. 2021, 11(16):7259.
- [3] Ling S J, Wang X L, Ding Y Y. Classification and Function of the Visual Variables of Static Map Symbol[J]. *Journal of Anhui Normal University(Natural Science)*, 2017, 40(1):69-76.
- [4] Bertin J. *Semiology of graphics: Diagrams, networks, maps* [M]. Madison, WI: University of Wisconsin Press, 1983.
- [5] Morrison J L. *A theoretical framework for cartographic generalization with the emphasis on the process of symbolization*. 1974.
- [6] MacEachren A M, *Visualizing Uncertain Information*. Cartographic Perspective, 1992,13, 10-19.
- [7] Mao W F. Research on validity of multiple visual variables in high-dimensional data visualization [D].Southeast University,2019
- [8] Ai B, Tang X H., Ai T H, Wang H B. Visualization of Spatio-Temporal Information Based on Transparency [J]. *Science of Wuhan University*,2012,37(02):229-232+259.
- [9] Huang L N, Li Y, Zhao W D. Application of Visual Variable in the Visualization of Campus Multiscale Temporal Information[J]. *Geospatial Information*, 2014,12(01):87-88+92+10.
- [10] Chen W , Pouliot J , Hubert F . How users perceive transparency in the 3D visualization of cadastre: testing its usability in an online questionnaire[J]. *GeoInformatica*, 2016, 21:1-20.
- [11] Fukiage T , Oishi T , Ikeuchi K . Reduction of contradictory partial occlusion in mixed reality by using characteristics of transparency perception[C]// *IEEE International Symposium on Mixed & Augmented Reality*. IEEE, 2012.
- [12] Ping J , Thomas B H , Baumeister J , et al. Effects of shading model and opacity on depth perception in optical see-through augmented reality[J]. *Journal of the Society for Information Display*, 2020.
- [13] Zhang L and Murdoch M J. Perceived Transparency in Optical See-Through Augmented Reality. 2021 *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2021, 115-120.
- [14] Borkin M A , Gajos K Z , Peters A , Mitsouras D, Melchionna S, Rybicki F J, Feldman C L, Pfister H. Evaluation of Artery Visualizations for Heart Disease Diagnosis[J]. *IEEE Transactions on Visualization and Computer Graphics*, 2011, 17(12):2479-2488.
- [15] Raimund Schnürer, Ritzi M , Ltekin A , et al. An empirical evaluation of three-dimensional pie charts with individually extruded sectors in a geovisualization context:[J]. *Information Visualization*, 2020, 19(3):183-206.
- [16] Dall'Acqua, L, Coltekin A , Noetzli J . A comparative user evaluation of six alternative permafrost visualizations for reading and interpreting temperature information[J]. *GeoViz_Hamburg* 2013, 2013.